

Indian Institute of Technology Dharwad
Electronics Club

Summer of Innovation 2026

BEYOND BITS

Design of a Compression System for Effective Communication

Important Information

Team Size: 1 – 4 Members
Registration Deadline: 7 June '26 (EOD)
Submission Deadline: 12 July '26 (EOD)
Registration Link: [Enter Beyond Bits](#)

Introduction

From streaming songs and sending photos to making phone calls and using GPS, modern communication systems constantly transmit massive amounts of digital data over limited bandwidth channels. Sending raw data directly is inefficient.

Designing integrated subsystems that combine data compression and reliable transmission is a key challenge in modern VLSI and communication system design.

Problem Statement

The objective is to design and implement an end-to-end system consisting of data compression, modulation, transmission, demodulation, and data reconstruction stages. The system should demonstrate an improvement in transmission efficiency through data compression while ensuring accurate recovery of the original data at the receiver.

System Architecture (Pipeline)

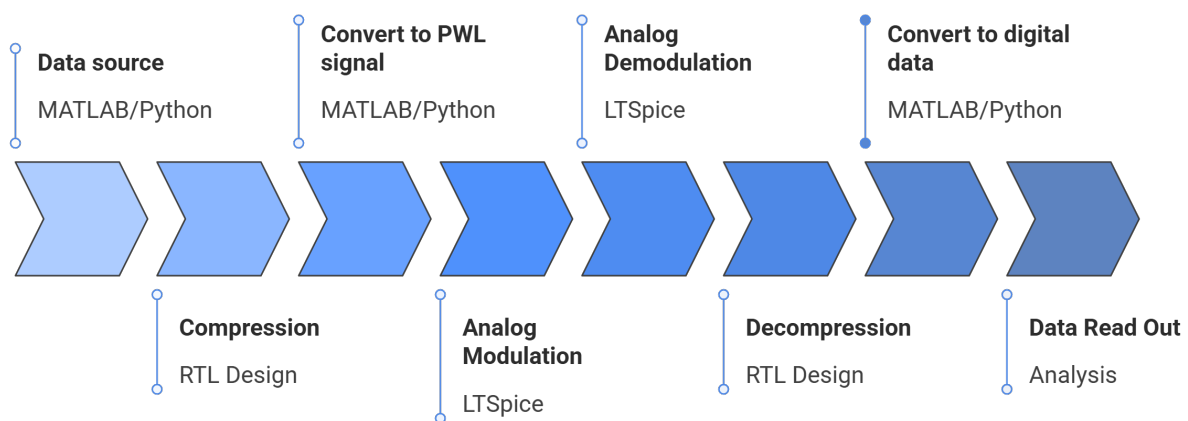


Figure 1: Baseline end-to-end pipeline.

Pipeline Enhancements

Enhance the existing pipeline by implementing **at least one** of the following three features:

- 1) Higher-order digital modulation schemes (e.g., M-ASK, M-FSK, where M represents the number of symbols)
- 2) Comparison and selection of efficient compression methods in RTL
 - Evaluate and compare at least three compression methods, and justify the selection of the final approach.
- 3) Addition of channel noise and RTL-based error-detection/error-correction coding.
 - Add noise to the signal during transmission using LTSpice or MATLAB/Python.

Design Approach

The project involves the design and integration of a simplified data transmission pipeline. Participants are expected to:

- Generate representative input data (text stream) using MATLAB or Python
- Study and select a suitable compression technique for RTL implementation (e.g., Run-Length Encoding or similar)
- Design and implement compression and decompression modules using Verilog/VHDL
- Generate PWL (Piecewise Linear) input files to use in LTSpice from compressed RTL output using MATLAB/Python
(The compressed digital output should be converted into a PWL waveform so that it can be used as an input source for LTSpice-based modulation)
- Design and simulate modulation and demodulation stages using LTSpice (ASK/FSK for the baseline design; higher-order schemes may be used as an enhancement).
- Convert demodulated output back into text/data format using MATLAB/Python for decompression
- Integrate all stages into a complete end-to-end simulation pipeline
- Evaluate system performance in terms of compression efficiency and successful reconstruction of data

Note: Make sure the selected enhancement is reflected in your design approach.

Evaluation Metrics

The solution will be evaluated based on:

- Design approach and system architecture
- Functional correctness
- Successful integration between RTL, MATLAB/Python and LTSpice environments, with additional credit for implementing using fundamental circuit-level components.
- Compression performance
- Innovation and design optimization
- Quality of implementation & evaluation of the selected enhancement.

Deliverables

Participants are expected to submit the following:

1. Technical Report containing:

- Problem understanding and design rationale
- System architecture and block diagrams
- Compression algorithm and its implementation
- Modulation and demodulation methodology
- System integration and verification
- Performance analysis including:
 - Compression ratio calculation:
$$CR = \frac{\text{Original Size}}{\text{Compressed Size}}$$
 - Verification of successful data demodulation and reconstruction
 - Relevant graphs, waveforms, and simulation results
 - Comparative analysis and justification of the selected compression method. Consider metrics such as compression ratio, implementation complexity, and resource utilization. (required only if the enhancement feature 2 is implemented)
 - BER and SNR analysis (required only if the enhancement feature 3 is implemented)

2. Files

- Verilog/VHDL source code for compression and decompression modules
- Testbenches for RTL verification
- MATLAB/Python scripts for input generation, PWL file generation & reconstruction, and Analysis
- LTSpice files for modulation and demodulation
- Simulation waveforms and output results
- Files related to the implemented enhancement(s)
- README file with instructions to run the complete system

Expected Outcome

The final system should demonstrate a functional end-to-end pipeline capable of efficient data compression, transmission using modulation techniques, and successful reconstruction of the original data after demodulation and decompression.

The system should be supported by quantitative analysis and simulation results.

Resources

- **Data Compression:**
[CMU Data Compression Docs](#) | [Run Length Encoding](#)
- **Digital Design:**
[Verilog Tutorials](#)
- **Communication Systems:**
[Fundamentals](#) | [Digital Modulation Techniques](#) | [ASK](#) | [FSK](#)
- **LTSpice:**
[Basics](#) | [ASK Implementation](#) | [FSK Implementation](#)
[Guide to use PWL File in LTSpice](#)
[Guide to add Noise & SNR calculation](#)
- **MATLAB:**
[MathWorks Documentation](#)

Tools/Software

- Verilog/VHDL simulation tools (ModelSim, Vivado, etc.)
- MATLAB
- Python
- LTSpice